

Compressive hereditary and buried disc disease

Part 2 | Compression: progression matters

Slow painless loss needs a trajectory, not a single acuity.

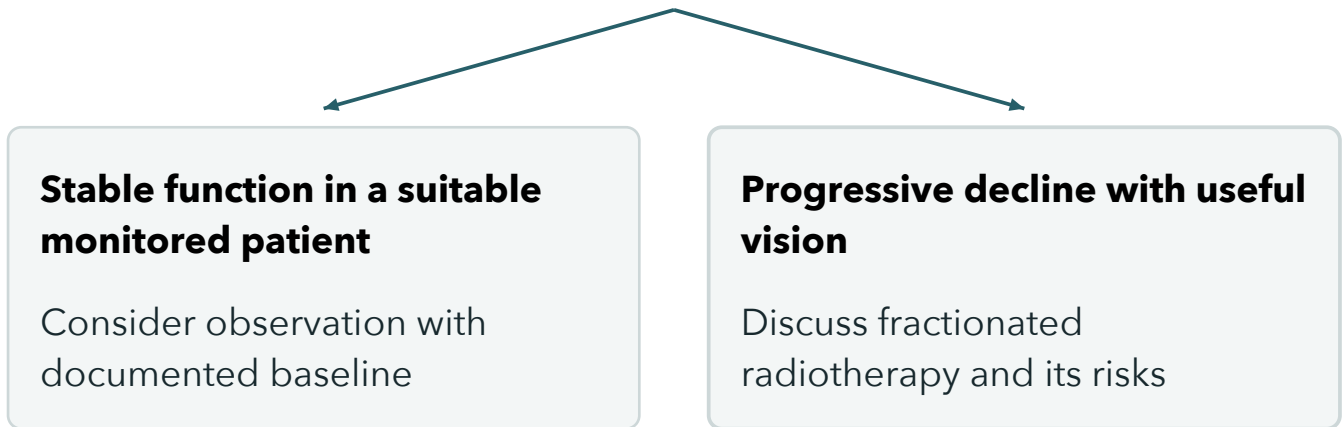
Finding	Implication
Progressive field or colour loss	Assess afferent dysfunction and targeted imaging
Disc pallor or collateral vessels	Localise with the full examination
MRI with appropriate orbital sequences	Define nerve, sheath and adjacent structures
Stable useful vision	Observation may be reasonable in selected disease

Establish the rate of change using comparable acuity, colour, fields and structural imaging. Compression may impair function without proptosis. Describe the anatomical relationship and remaining vision before proposing treatment. The management question is whether intervention protects useful function sufficiently to justify its risks.

ONSM: preserve function when possible

Progressive disease with useful sight can favour radiotherapy.

How is an optic nerve sheath meningioma behaving?



Direct excision can threaten the optic nerve and its blood supply. The choice should account for function, growth, location and extension, rather than treating a benign histological label as reassurance. Explain why control of growth and preservation of vision are separate outcomes; longstanding loss may not reverse even when the lesion is controlled.

Buried drusen and PHOMS

A structural clue must not cancel an intracranial-pressure assessment.

PHOMS: peripapillary

True drusen: within disc



Schematic section. PHOMS are not synonymous with buried drusen.

Original section: true drusen lie within the disc; PHOMS are peripapillary.

Use enhanced-depth imaging and complementary tests to characterise a suspected deposit. PHOMS represent a different structural finding and occur with several disc states; they are not synonymous with drusen. If symptoms or examination suggest raised pressure, continue the appropriate work-up even when a bright peripapillary structure is visible on OCT.

Inherited optic neuropathy: expression is variable

The same mitochondrial variant does not predict identical disease.

Finding	Implication
Maternal pedigree	Consider mitochondrial transmission
Heteroplasmy when present	Different mutant fractions can influence expression
Incomplete penetrance	An unaffected carrier remains possible
Counselling	Include sex, environment and molecular context

Distinguish transmission of the variant from expression of the phenotype. A variable clinical picture does not turn a mitochondrial disorder into an autosomal one. Explain the limits of predicting onset and severity from carriage alone, and avoid reassuring a relative simply because another carrier has remained well. Specialist genetic counselling should accompany confirmed disease.

Disc swelling has more than one mechanism

Use tempo and associated signs to challenge the initial label.

Finding	Implication
Abrupt painless loss	Consider an ischaemic optic neuropathy
Inflammation or central signs	Broaden the neurological differential
Vitreous attachment deforming the disc	Consider vitreopapillary traction
Progressive compression	Target orbital and intracranial imaging

In a viva, identify the feature that contradicts the first diagnosis. A swollen disc and poor vision can arise from different vascular, mechanical and inflammatory processes. Relate the history to OCT and imaging rather than counting shared signs. The next investigation should answer a specific unresolved mechanism and change the treatment or surveillance plan.